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# Java Project

## Third Review

# Student Course Registration System

**Presented by**


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# ABSTRACT


1. Java-based application for automating course registration

2. Replaces manual and error-prone registration methods

3. Allows students to register, update, view, and delete courses

4. Prevents duplicate registrations using validation

5. Ensures accurate and secure data management

6. Provides user-friendly interface using Java Swing

7. Stores data in MySQL / Oracle database

8. Reduces time consumption and human errors

9. Improves efficiency in managing student and course records

10. Supports real-time data access and updates

# INTRODUCTION

- Educational institutions require efficient systems to manage course registrations.
- Traditional manual methods are:
  - Time-consuming
  - Error-prone
  - Difficult to manage and update
- Student Course Registration System (SCRS) automates the registration process
- Developed using Java programming and database integration
- It allows students to:
  - Register for courses
  - View registered courses
  - Update course details
  - Delete course registrations
- Provides:
  - Fast and accurate registration
  - Secure data storage
  - Easy access to student and course information

- Improves overall **efficiency, accuracy, and management**

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# PROBLEM STATEMENT

- Manual course registration systems face several challenges:
    - Data redundancy and duplication
    - Errors in course registration
    - Difficulty in tracking registered courses
    - Time-consuming registration process
- Lack of proper data management
- Lack of a centralized system leads to inefficiency
- There is a need for a **reliable, automated system** that:
    - Manages student and course records
    - Allows easy course registration
    - Prevents duplicate registrations
    - Enables quick updates and deletions
    - Provides accurate and secure data storage

# SCOPE OF THE PROJECT

## SCOPE OF THE PROJECT:

- Suitable for colleges and universities
- Manages student course registration efficiently
- Stores student and course information securely
- Allows adding, updating, and deleting course registrations
- Prevents duplicate registrations
- Provides quick access to registered course details
- Ensures accurate and organized data management

**Future scope:**

- Online web-based registration system
- Mobile application integration
- Role-based login

# MODULES

**Admin Dashboard**

- Central control panel
- Overview of students and courses

**Student Management**

- Add, update, delete student details
- Maintain student records

**Course Management**

- Add, update, delete courses
- Manage course information
- Track course availability

**Key Features:**

- Ensures data consistency and integrity
- Uses primary and foreign keys for relationships
- Prevents duplicate registrations
- Supports secure and efficient data handling

# DATABASE DESIGN

•The system uses a **relational database (MySQL/Oracle)** for data storage
• Multiple tables are used to manage student and course information efficiently

## Tables Used:

### Student Table

- student_id (Primary Key)
- name
- roll_no (Unique)
- email
- department

### Teacher Table

- emp_id (Primary Key)
- name
- password

### Course Table

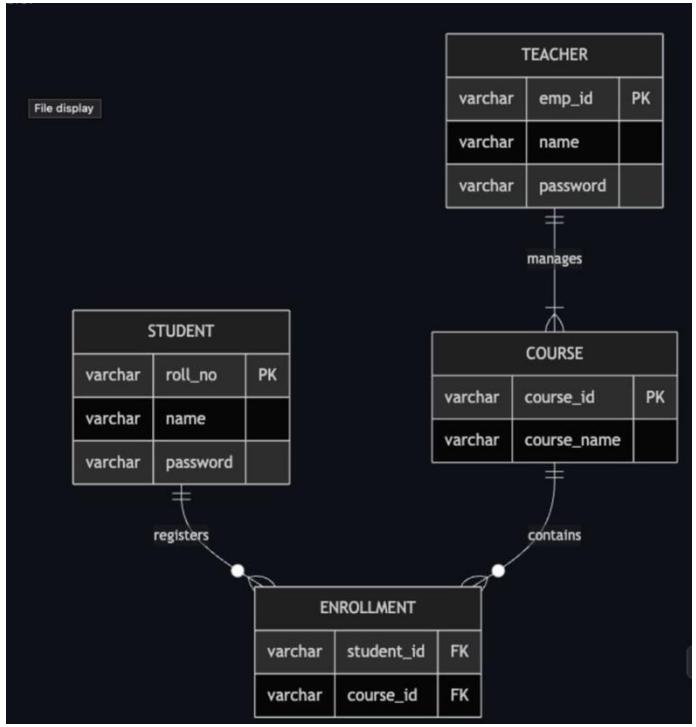
- course_id (Primary Key)
- course_name
- course_code (Unique)
- credits

### Enrollment Table

- student_id (Foreign Key)
- course_id (Foreign Key)

| ENROLLMENT | | |
|---|---|---|
| varchar | student_id | FK |
| varchar | course_id | FK |

| COURSE | | |
|---|---|---|
| varchar | course_id | PK |
| varchar | course_name | |

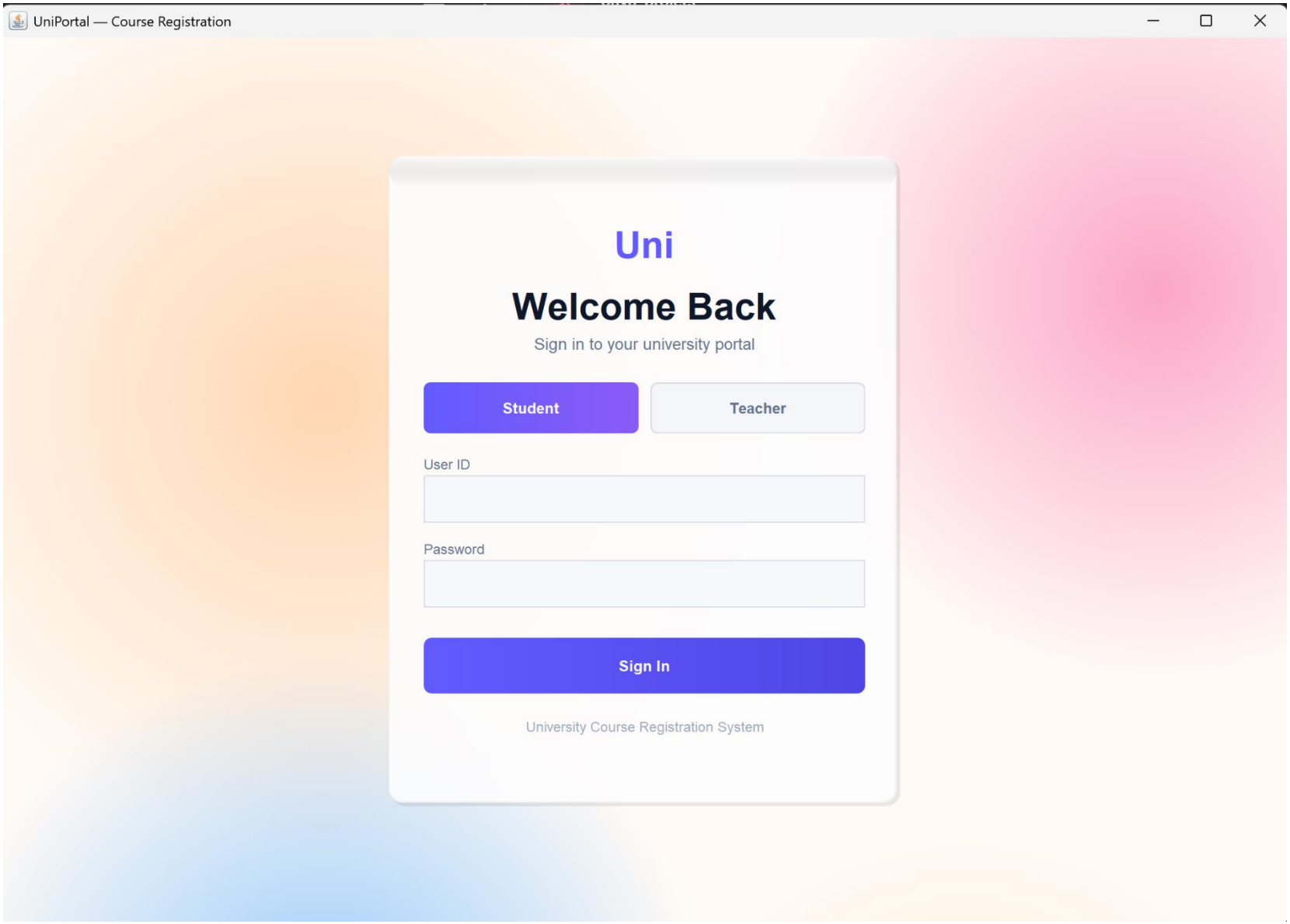



| STUDENT | | |
|---|---|---|
| varchar | roll_no | PK |
| varchar | name | |
| varchar | password | |

| TEACHER | | |
|---|---|---|
| varchar | emp_id | PK |
| varchar | name | |
| varchar | password | |

# OUTPUT SCREENS

## Output Screens




**University portal Login Page**

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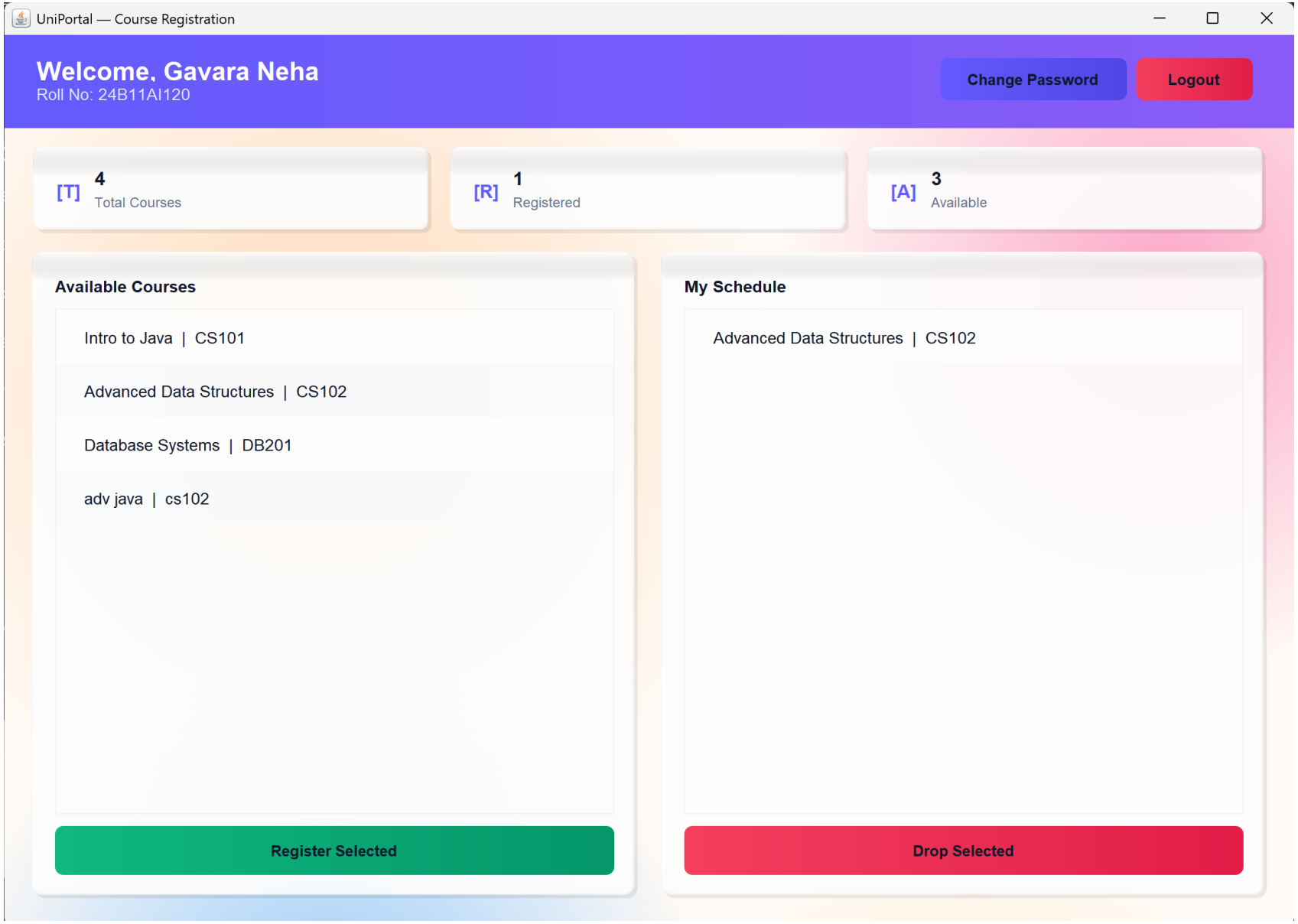
# OUTPUT SCREENS

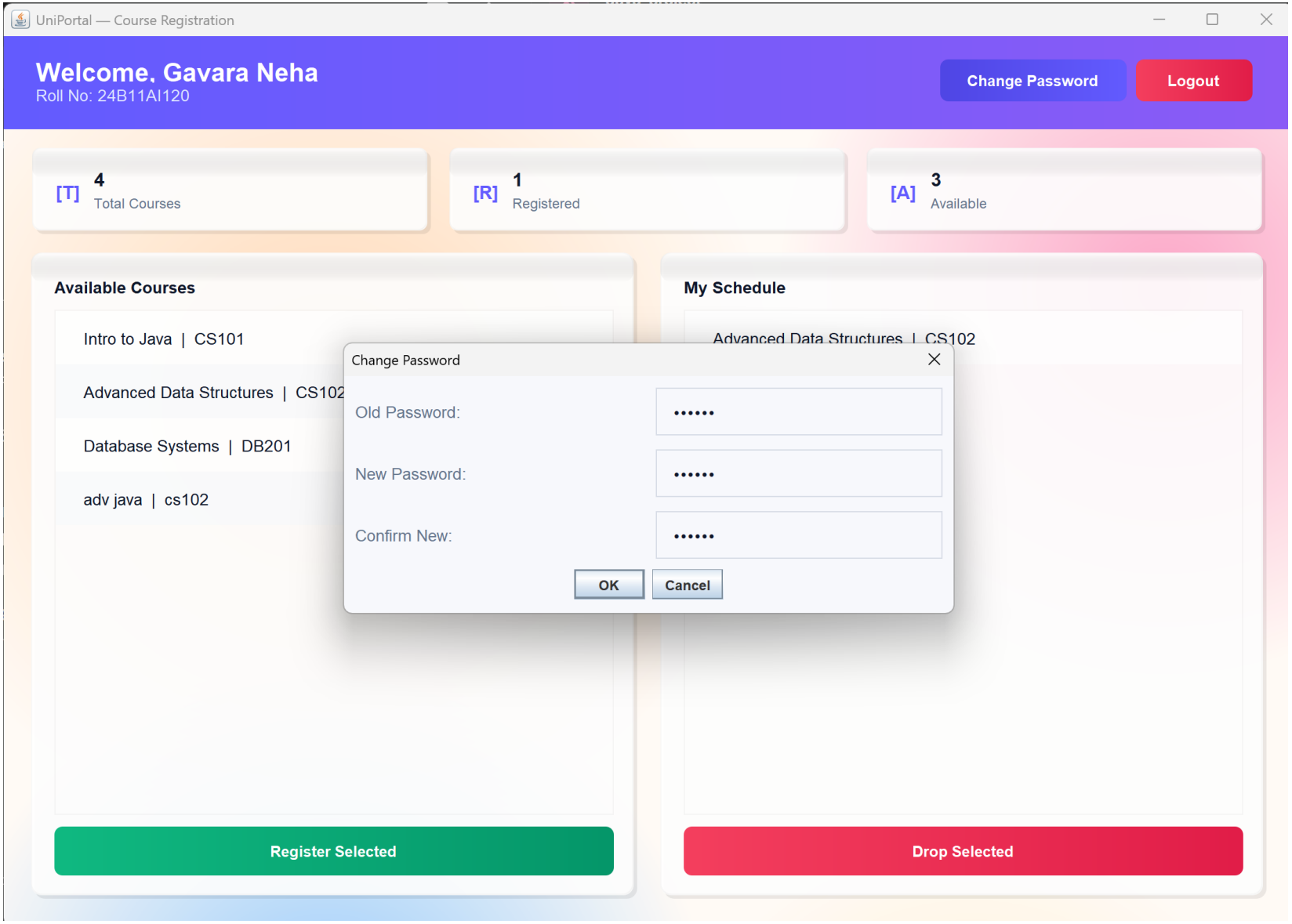
## Output Screens




**Student Dashboard [ Course Registration ]**

# OUTPUT SCREENS

## Output Screens




**To Change password**

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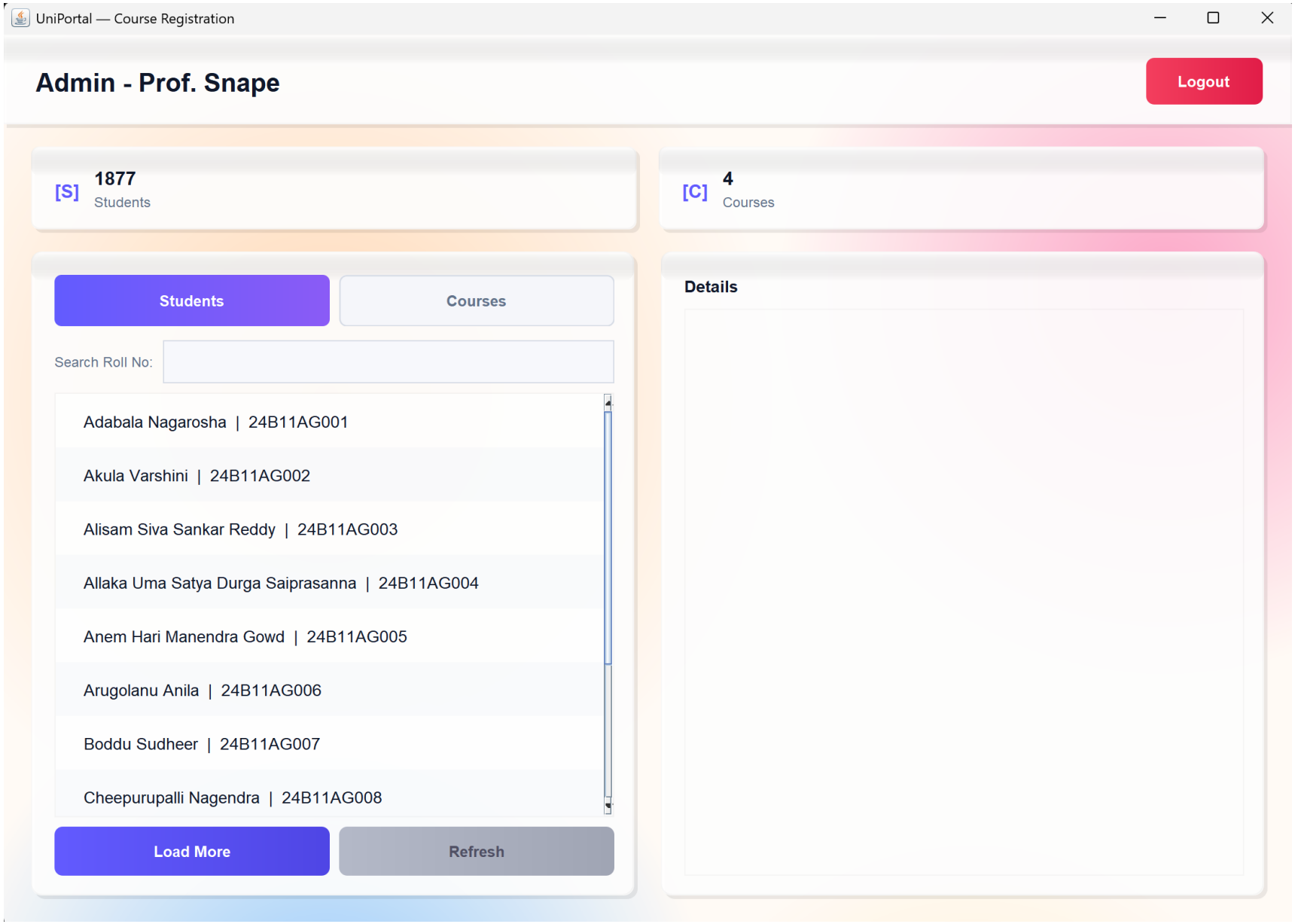
# OUTPUT SCREENS

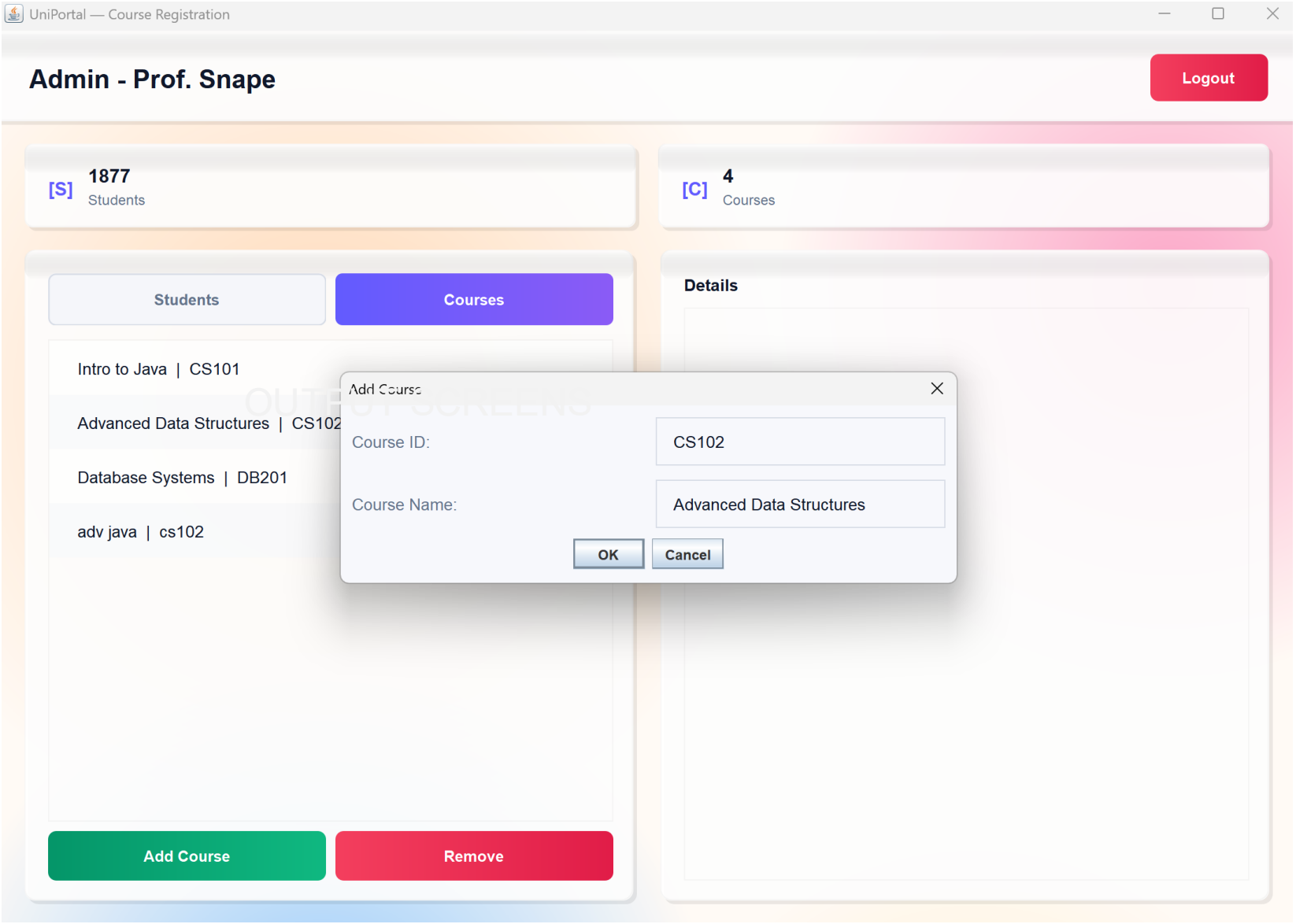
## Output Screens




**Teacher /Admin Dashboard**

# OUTPUT SCREENS

# Output Screens




**Teacher /Admin Dashboard Course selection**

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# CONCLUSION

- The Student Course Registration System provides a centralized platform for managing student and course enrollments

- It successfully automates the course registration process, reducing manual effort and errors

- The system supports complete CRUD operations for students, courses, and registrations

- Role-based access ensures secure usage for both admin and students

- Integration of Java Swing, Hibernate (JPA), and MySQL ensures efficient data handling and persistence

- The application demonstrates a structured layered architecture for scalable development

- Overall, the system improves efficiency, accuracy, and user experience in academic course management

# THANK YOU